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Bayesian model reconstruction based on spectral line observations with pomme

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Summary

A typical problem in astronomy is that, for most of our observations, we are restricted to the plane of the sky. As a result, these observations are always mere projections containing only partial information about the observed object. Luckily, some frequency bands of the electromagnetic spectrum are optically thin, such that we receive radiation from the entire medium along the line of sight. This means that, at least in principle, from the observed radiation, we can extract information about the physical and chemical conditions along the entire line of sight. This is especially the case for spectral line radiation caused by transitions between the quantized energy levels of atoms and molecules. Rotational transition lines are particularly interesting, since they are excited in many astrophysical environments and can easily be resolved individually. Using spectral line observations, we can infer information about physical and chemical parameters, such as abundance of certain molecules, velocity, and temperature distributions. To facilitate this, we built a Python package, called pomme, that helps users reconstruct 1D spherically symmetric or generic 3D models, based on astronomical spectral line observations. A detailed description and validation of the methods can be found in (De Ceuster et al., 2024).

Statement of need

Spectral line observations are indispensible in astronomy. As a result, many line radiation transport solvers exist to solve the (forward) problem of determining what spectral line observations of a certain model would look like Matsumoto et al. (2023). However, far fewer tools exist to efficiently sovle the more pressing inverse problem of determining what model could correspond to certain observations. Although one can use existing forward solvers to iteratively solve the inverse problem, the resulting process is often slow, cumbersome, and leaves, much room for improvement Danilovich et al. (2024). Therefore, in pomme, we have implemented the forwad problem of line radiative transfer in the PyTorch framework (Paszke et al., 2019) to leverage the autograd functionality (Paszke et al., 2017), which allows us to efficiently solve the inverse problem in a streamlined way.

Example

Figure 1 shows an application of pomme, where it was used to reconstruct the abundance distribution of NaCl (table salt) around the evolved star IK Tauri. The reconstruction is based

on observations of the NaCl ($J = 26 - 25$) rotational line, taken with the Atacama Large (sub)Millimetre Array (ALMA). The original analysis was done by Coenegrachts et al. (2023), and we improved their methods using pomme as described in De Ceuster et al. (2024).

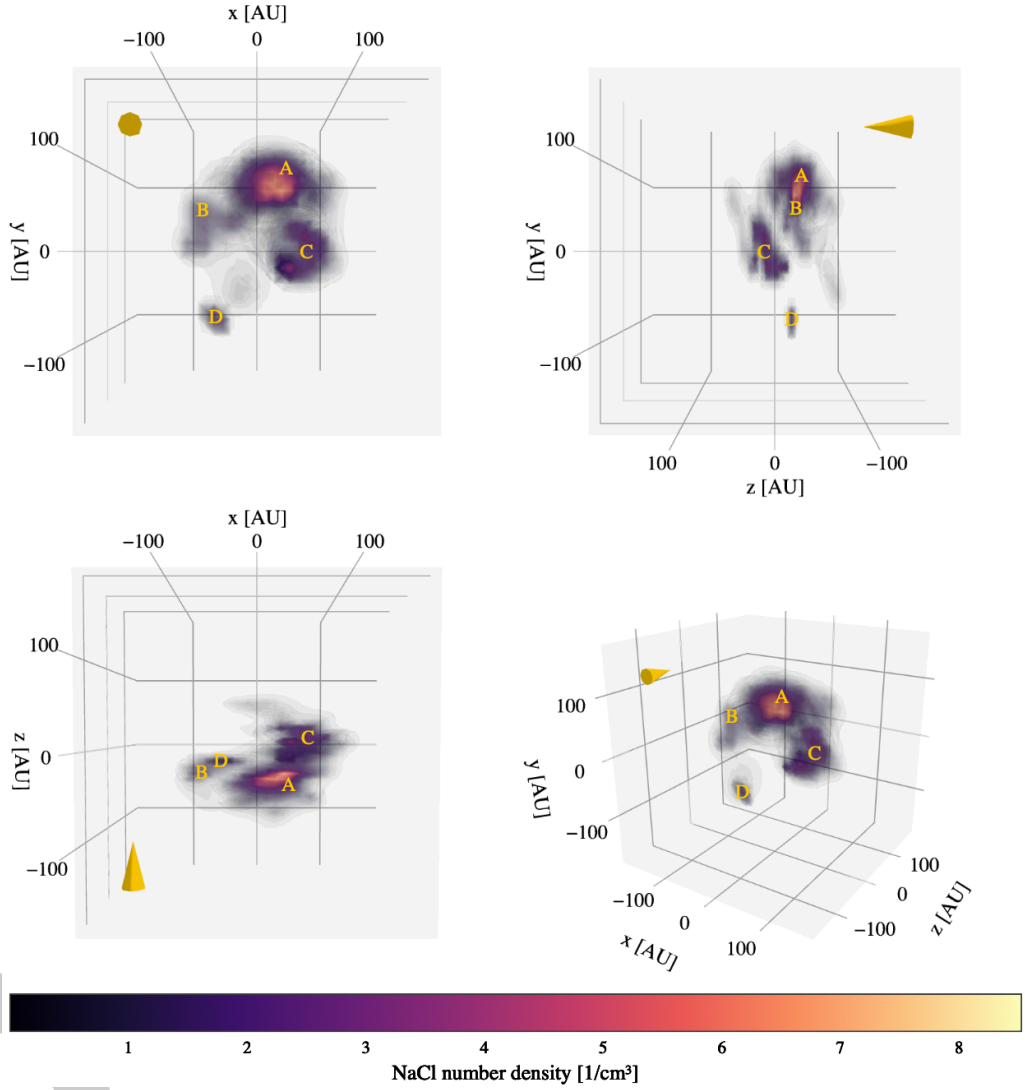

Figure 1: Reconstruction of the NaCl abundance distribution around the evolved star IK Tauri, created with pomme. An interactive version of the figure is available in the documentation.

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